YAKEEN

NEET 2.0 2027

Physics | Lecture - 01

Motion in a Plane

By- Manish Raj (MR Sir)



H/w goal

→ Check today class note & do all Practice given here

Todays lecture

Juggles Prob^m air resistance

Projectile 2-D motion

Sir ji aj apne Jo padhaya bilkul maja agaya . Aj tak kabhi bhi aise nahi padha sir ji. 11th me ye sab question ko ratta marte the . Kichu samajh nahi ata tha . Lekin aj hum khud qsn kar parahe hai. Ap mahan ho sir ji .
Aap physics ka king ho ❤️

Double-tap to ❤️

Message...

Edited

Phycics from you and was like woww is this that much easy i was so happy that the fear turned into happiness i literally before sitting to class 1 touches your feet and i m able to solve all those numericals which i was afraid of solving and its just because of you sir thank yoh soo soo soo much 🙏😭🥹 i dont know whats gonna happen next year i dont know whats my fate but sir i learned to work hard , learned to sit for study for hours sirr thankk youu thank youuu 🥹
Sir lots of love .
I dont you whether you are going to see this message or not but whatever you will read this , this are all true feelings with whole heart 🙏

You don't follow each other on Instagram

View profile

8:08 PM

Sir aapne jo experiment btaya tha paper ko fall kra kr pakdne vala aaj mene vo krvaya apne ghr valo se sir mene 3500 rupees kama liye bhot achi trick h sirMR STAR OP 👊✨
✌🏻

10 % Mujhe de do

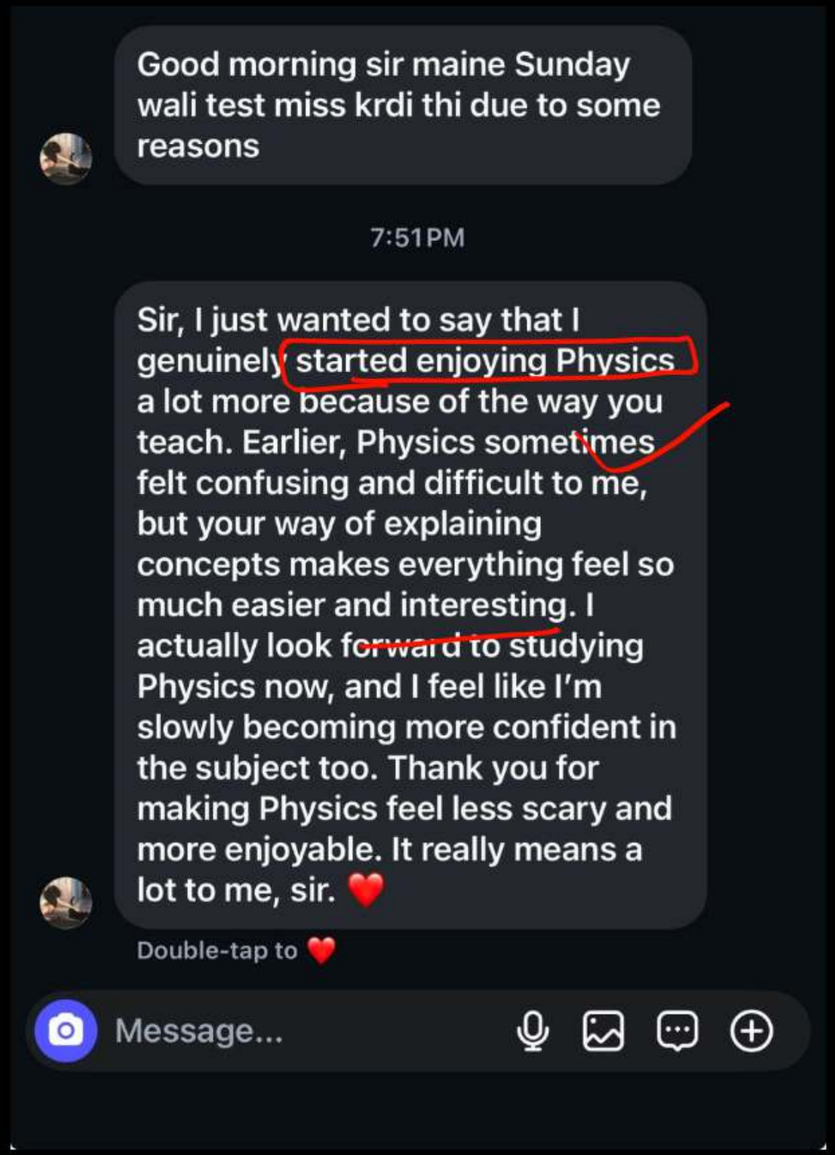
New messages

Edited

Sir credit to aap 100 % le lijiye pr vo thode se paise garib k liye chhod dijiye 😅

Double-tap to ❤️

Message...

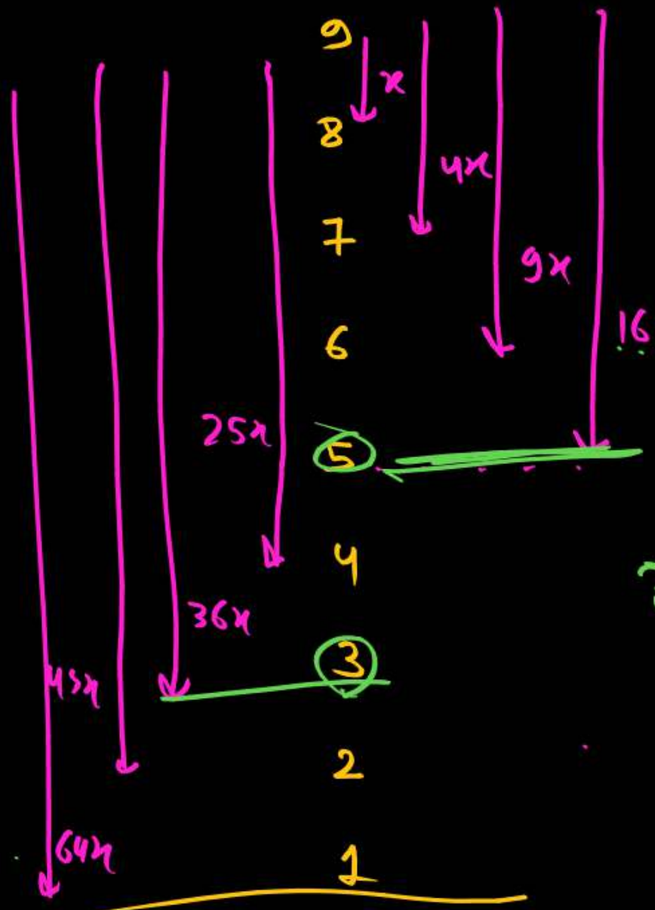

Good morning sir maine Sunday wali test miss krdi thi due to some reasons
7:51 PM
Sir, I just wanted to say that I genuinely started enjoying Physics a lot more because of the way you teach. Earlier, Physics sometimes felt confusing and difficult to me, but your way of explaining concepts makes everything feel so much easier and interesting. I actually look forward to studying Physics now, and I feel like I'm slowly becoming more confident in the subject too. Thank you for making Physics feel less scary and more enjoyable. It really means a lot to me, sir. ❤️
Double-tap to ❤️
Message...

Question

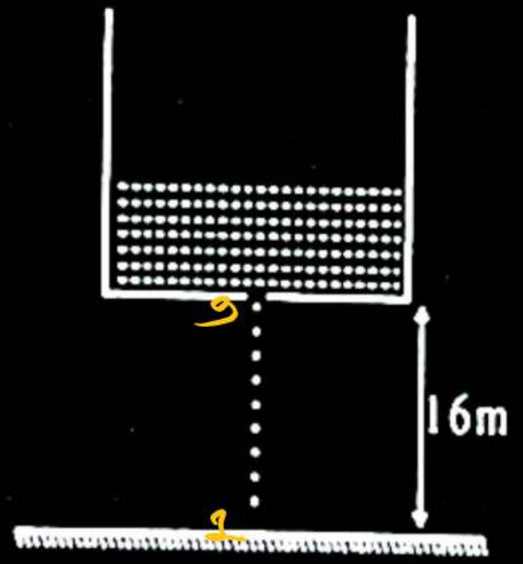
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Water drops fall with negligible velocity at regular intervals from a hole at the bottom of a vessel placed 16m from the ground. The ninth drop is about to fall when the first drop just falls on the floor. Find the distance between the third and fifth drop at this instant in meters.

1. 2
2. 5
3. 8
4. 9

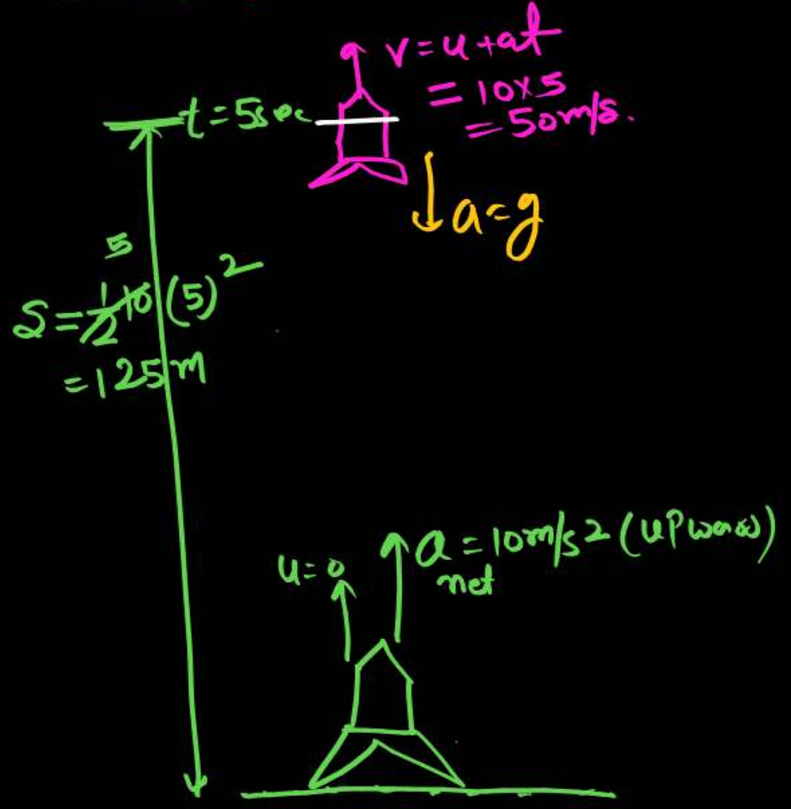


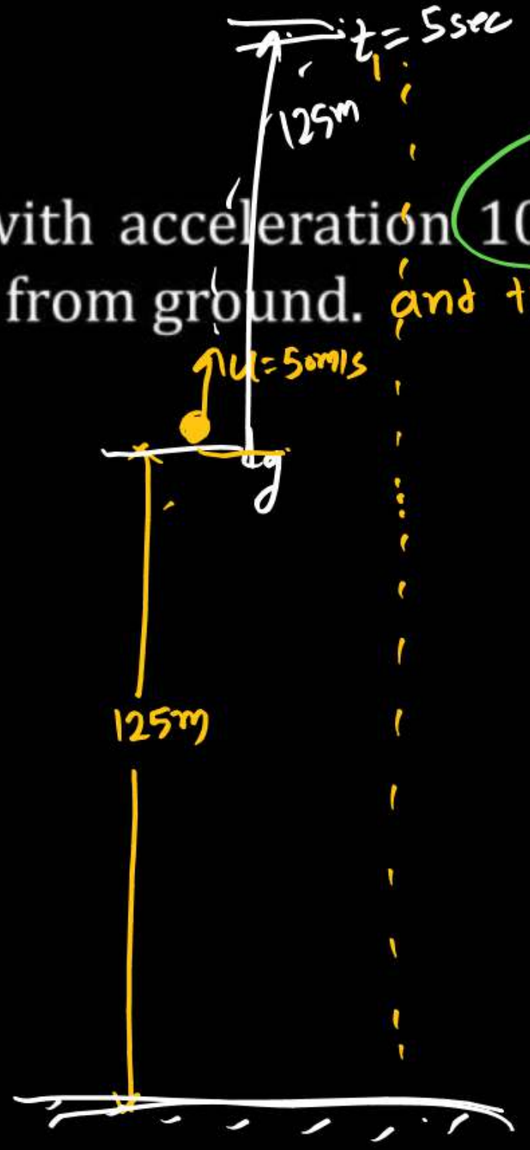
$36x - 16x = 20x$



Question

Rocket starts his motion in upward direction with acceleration 10 m/s² upward. After 5 sec engine off then find maximum height from ground. and time of flight.





$H_{\text{from ground}} = 250\text{ m}$

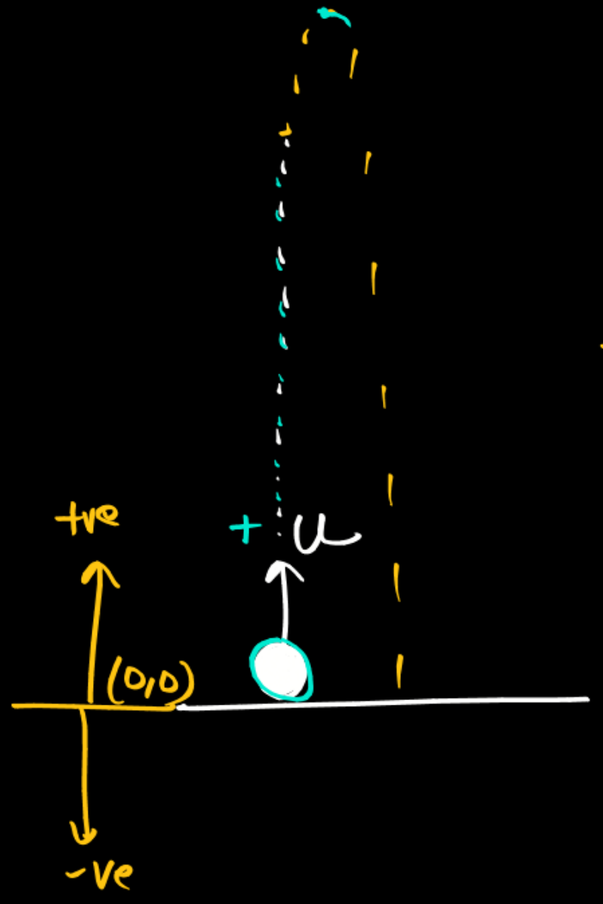
$T_f = (5\text{sec} + 7.1\text{sec})$ after engn off

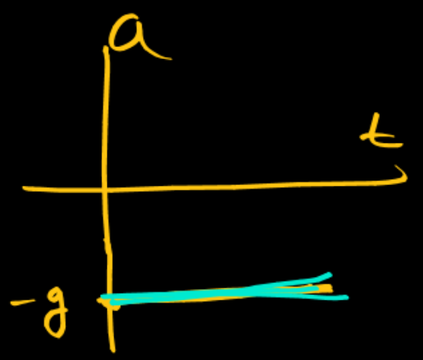
$T = \sqrt{\frac{2\times 250}{g}}$

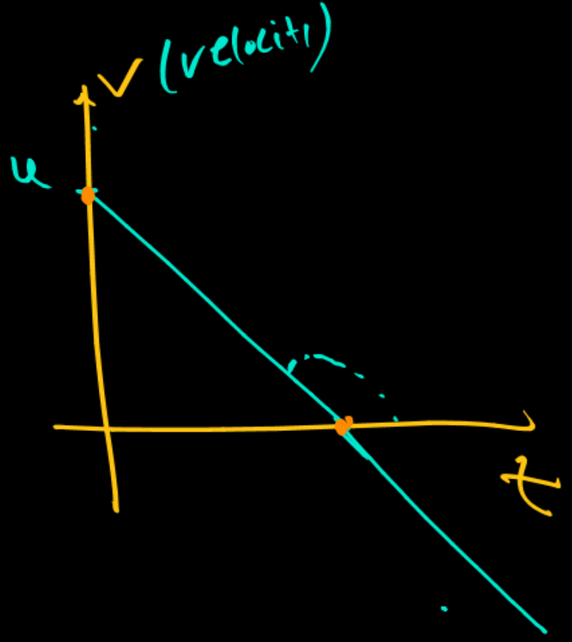
$= \sqrt{50}$

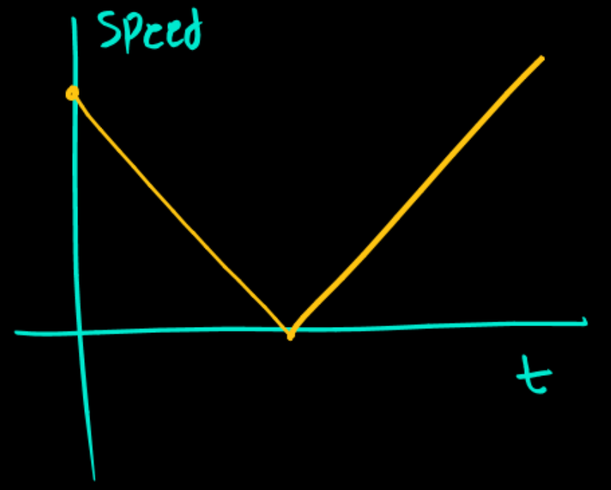
$= 7.1\text{sec}$

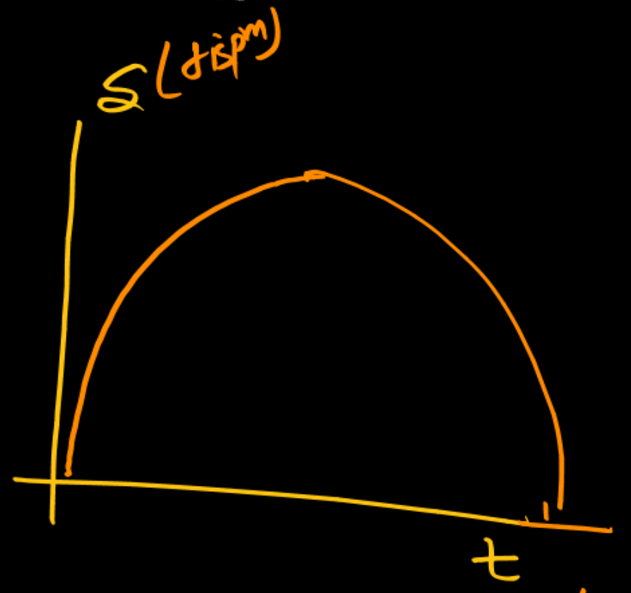
② Ball is projected up from ground, Draw x/t, v/t, a/t graph.

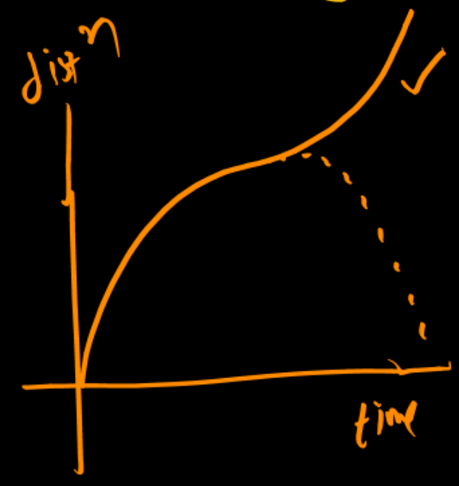




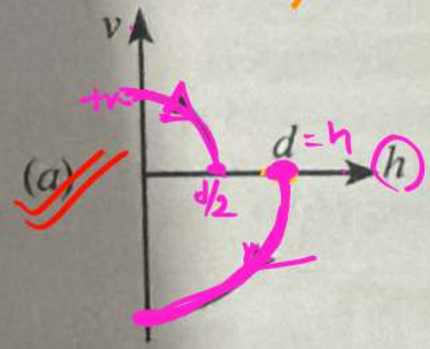


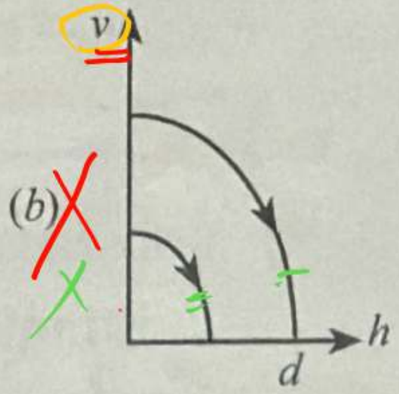


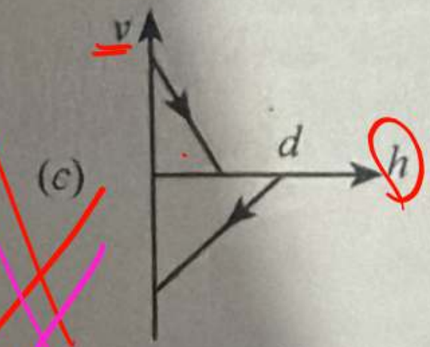


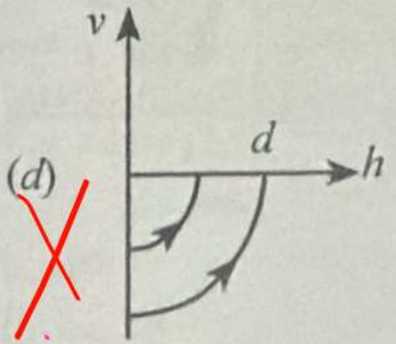


13. A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height $d/2$. Neglecting subsequent motion and air resistance, its velocity v varies with height h above the ground as **(IIT-JEE 2000)**

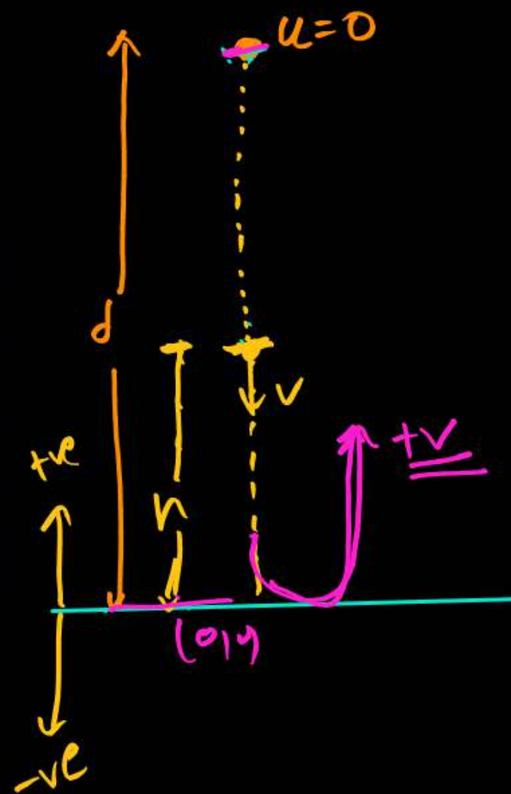








14. A particle P is sliding down a frictionless hemispherical bowl. It



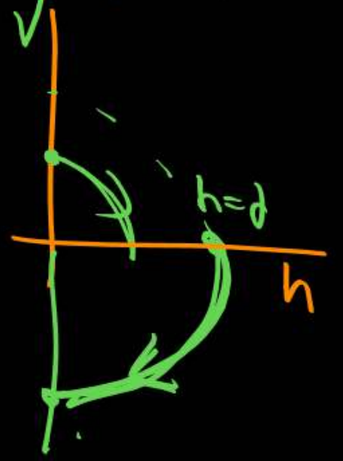
$v^2 - u^2 = 2as$

$v = \sqrt{2as}$

$= \sqrt{2(+g)(+(d-h))}$

$v = \sqrt{2g(d-h)}$

$v = \sqrt{2g(d-h)}$



Q Jugglar wants to keep n-Ball in air, if he throw each Ball with Speed u, then find time intraval b/w any two Ball.

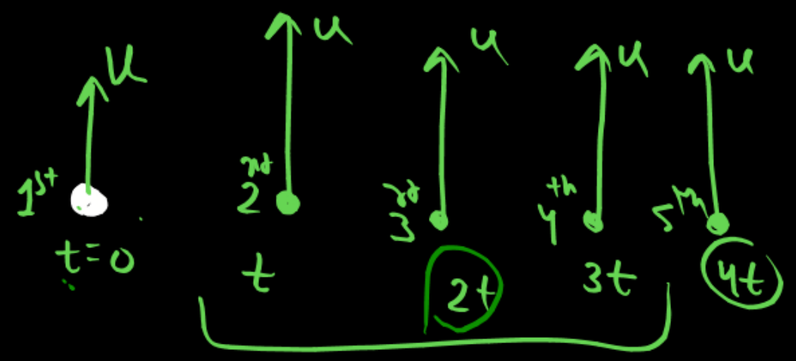
- u = intial velocity of each ball = same
- t = time gap same b/w each ball

he want to keep 4-Ball in air ✓

$$t_{1st\ Ball} = \frac{2u}{g} = 4t$$

if he want to keep n-Ball in air

$$\frac{2u}{g} = nt$$



Jugglar wants to keep 5 - Ball in air, with time intraval 2-sec then find intial speed of ProTection ??

Soln

$$\frac{2u}{g} = 5t$$

$$\frac{2 \times u}{10} = 5 \times 2$$

$$u = 50 \text{ m/s} \checkmark$$

Juggler wants to keep 10 Ball in air, each Ball rise upto 80m height then find time gap B/w Balls.

$t = ??$ $\quad n = 10$

$h_{max} = 80\,m$

$h_{max} = \frac{u^2}{2g}$

$80 = \frac{u^2}{2 \times 10}$

$\sqrt{1600} = u$

$u = 40$

$\frac{2u}{g} = nt$

$\frac{2 \times 40}{10} = 10\,t$

$t = 0.8\,sec$



$$T = \sqrt{T_1 T_2}$$

Try to prove it //

Question

A man throws ball with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time? (Given $g = 9.8\ m/s^2$) **[MR* (2003)]**

1. More than 19.6 m/s
2. At least 9.8 m/s
3. Any speed less than 19.6 m/s
4. Only with speed 19.6 m/s

Question

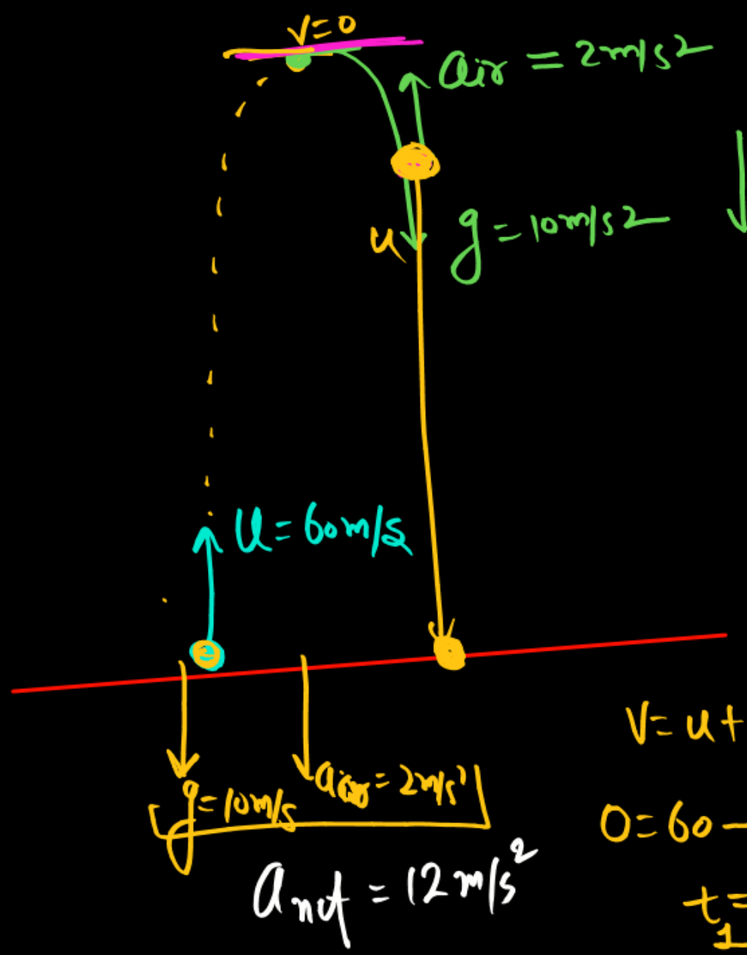
A person sitting in the ground floor of a building notices through the window, of height 1.5 m, a ball dropped from the roof of the building crosses the window in 0.1 s. What is the velocity of the ball when it is at the topmost point of the window? ($g = 10\ m/s^2$)

(2020-Covid)

1. 14.5 m/s
2. 4.5 m/s
3. 20 m/s
4. 15.5 m/s

$v = \sqrt{2gh}$

(#) Ball is projected up with speed 60m/s. and constant air friction $2m/s^2$ acting opposite to the dirn of motion then find H_{max}, T_{up} T_{down}.



$V = u + at$

$0 = 60 - 12t$

$t_1 = 5sec$ (Upward)

$H_{max} = \frac{u^2}{2g_{eff}}$

$= \frac{60 \times 60^5}{2 \times 12}$

$= 150m$

Downward motion

$s = ut + \frac{1}{2}at^2$

$150 = \frac{1}{2} \times 8 \times t^2$

$t_2 = \sqrt{\frac{150}{4}} = \frac{12.2}{2}$

$t_2 \simeq 6.12 sec$

$t_{up} < t_{down}$

$u_{collis} = \sqrt{2 \times (8) 150}$

$= \sqrt{300 \times 8}$

$= 48.99 m/s$

MR* Box for air friction in motion under gravity.

(#) $u_{Project} > u_{collision}$.

(#) $t_{up} < t_{down}$

Motion in Plane.

Motion under gravity ke Bad hi ye chapter Padhna.

① Position of object $\vec{r} = t\hat{i} + 3t\hat{j}$ then Motion of object

MR scan

is :-

$\vec{r} = x\hat{i} + y\hat{j}$

(a) 1-D (straight line) (181.)

(b) 2-D [curved Path]

(c) Can't say

$x = t \quad -(i)$

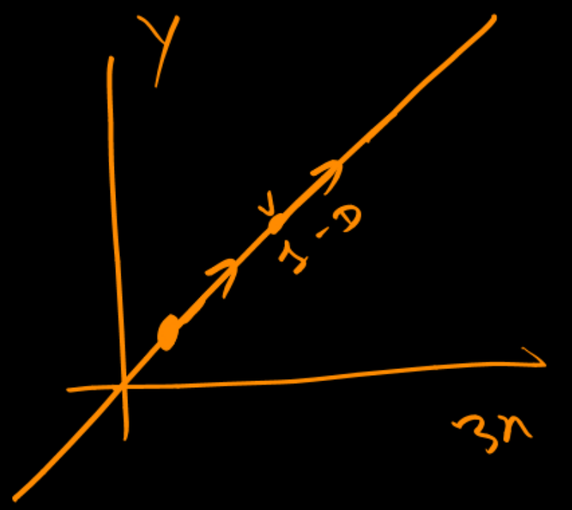
$y = 3t \quad -(ii)$

eqn of Trajectory

Path ka eqn

$\frac{x}{y} = \frac{1}{3}$

$y = 3x$



Cont. ② Position of object in x/y plane $\vec{r} = 4\sin(t)\,\hat{i} + 3\sin(t)\,\hat{j}$

then object is moving on :-

(a) Curved Path 28-1.

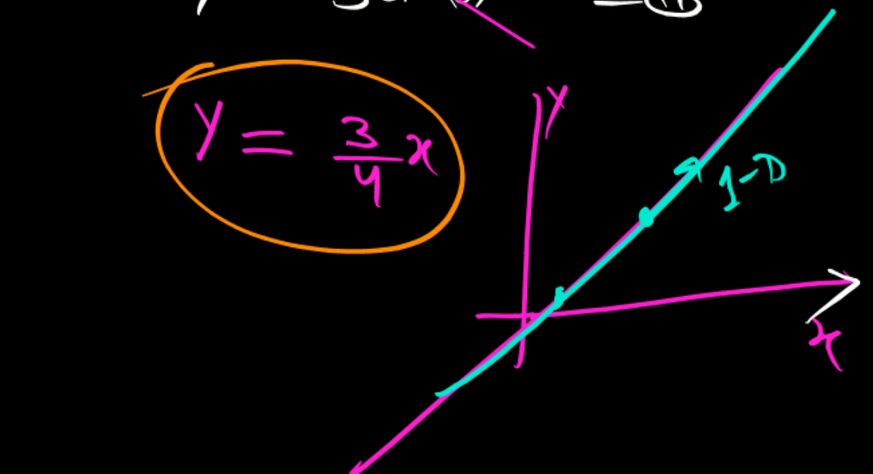
(b) ~~Straight line~~ (1-D) 57-1.

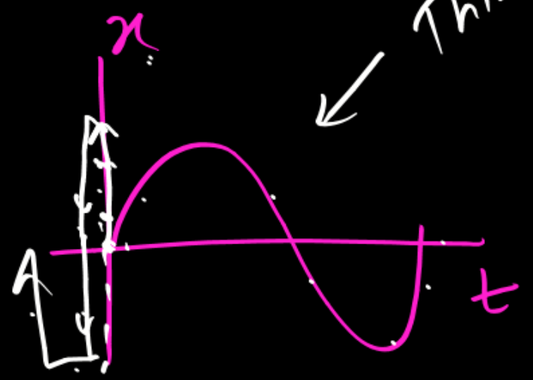
(c) Can't say.

$x = 4\sin t \quad -(i)$

$y = 3\sin(t) \quad -(ii)$

$y = \frac{3}{4}x$





This is not Path

Copy

MR* Box → for equation of Path.

- equation of x/t, v/t, a/t or their graph does not represent Path of object.
- only x/y ka relation s x/y ka graph Path ko represent karta hai
- Eqn of Trajectory ke liye x s Y ka relation find.

③ If $x = 3t$ ① & $y = 2t^2$ ⑪ then eqn of Path (Trajectory)

Soln

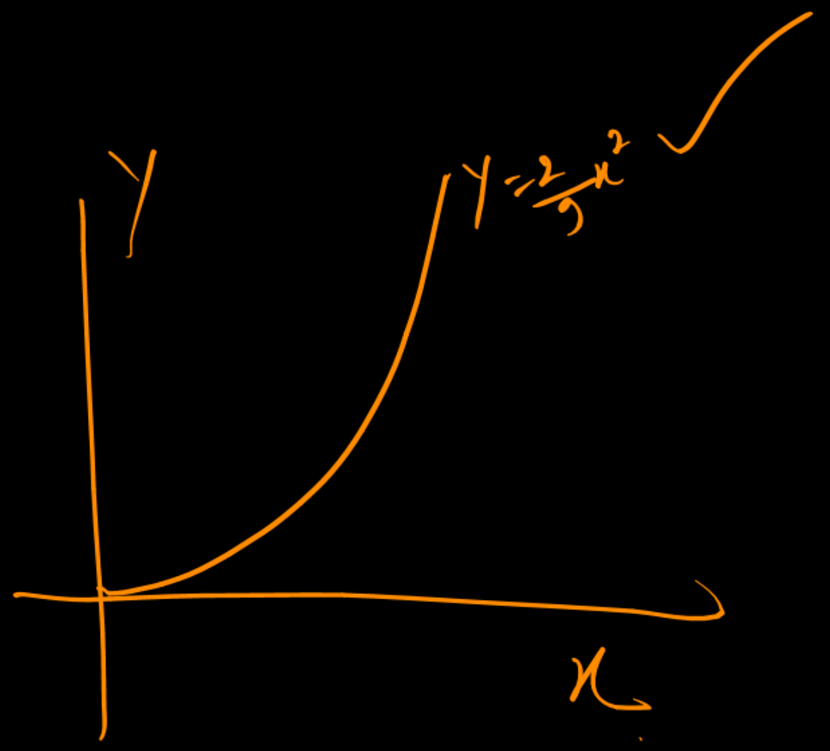
Put value of t in eq ⑪ from ①

$t = x/3$

$y = 2(t)^2$

$y = 2\left(\frac{x}{3}\right)^2$

$y = \frac{2x^2}{9}$ → Parabola



④ If $\vec{r} = 2\sin(t)\,\hat{i} + 2\cos(t)\,\hat{j}$ then eqn of Trajectory

sirf tuka bolde (-1)

(a) 1-D (straight line) (43%)

(b) 2-D (curved) (54%)

(open add)

copy

wrong (Not copy)

$x = 2\sin(t)$

$y = 2\cos(t)$

$\frac{x}{y} = \tan(t)$

$y = x\cot(t)$

$x = 2\sin t$ — ①

$y = 2\cos t$ — ②

$x^2 + y^2 = 4\sin^2 t + 4\cos^2 t$ ($①^2 + ②^2$)

$x^2 + y^2 = 4(\sin^2 t + \cos^2 t)$

$x^2 + y^2 = 4$ → Object is moving on circle ✓

⑤ If $\vec{r} = 2\sin(t)\,\hat{i} + 4\cos(t)\,\hat{j}$ then eqn of Trajectory

$\vec{r} = x\hat{i} + y\hat{j}$

Soln

$x = 2\sin t$

$y = 4\cos t$

$\frac{x}{2} = \sin t \quad -(i)$

$\frac{y}{4} = \cos t \quad -(ii)$

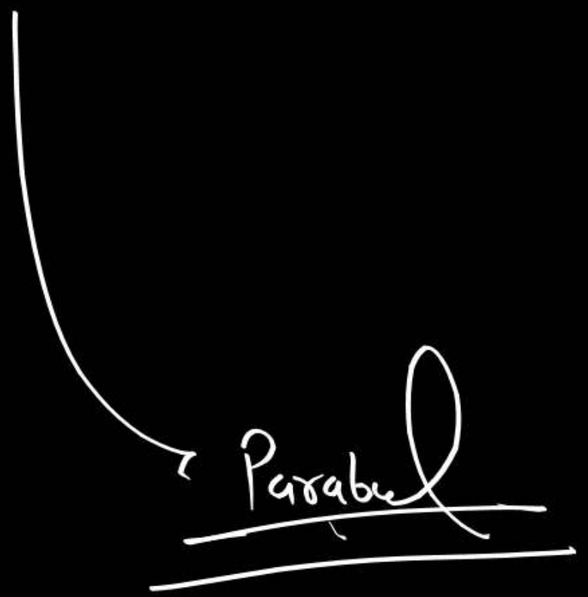
$$\frac{x^2}{4} + \frac{y^2}{16} = \sin^2 t + \cos^2 t = 1$$

→ Path is elliptical

Question

No copy

Position of object at time 't' $\vec{r} = 2\,t\,\hat{i} + 4\,t^2\,\hat{j}$, then find equation of trajectory.



Question 6

Velocity of object $\vec{V} = 2\,\hat{i} + x\,\hat{j}$ then equation of trajectory.??

$\vec{V} = v_x\hat{i} + v_y\hat{j}$

$\vec{V} = 2\hat{i} + x\hat{j}$

$\vec{V_x} = 2 \qquad \vec{V_y} = x$

$\frac{dx}{dt} = 2 \qquad \frac{dy}{dt} = x$

$\int dx = \int 2\,dt$

$x = 2t \quad -(1)$

$dy = x\,dt$

$\int dy = \int 2t\,dt$

$y = 2\left(\frac{t^2}{2}\right) = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4}$ Ans

MP*

$\frac{v_x}{v_y} = \frac{2}{x}$

$\frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{2}{x}$

$\int x\,dx = \int 2\,dy$

$\frac{x^2}{2} = 2y$

$y = \frac{x^2}{4}$

Question

(7) HW

A particle moving with velocity $V = y\,\hat{i} + x\,\hat{j}$ then find equation of trajectory.

[IIT-2020]

Motion in Plane [2-D] ⟹ $[1\text{-}D]_x + [1\text{-}D]_y$

2-D motion ⟶ (1-D motion) + (concept of vector)

* 2-D ka koi eqn of motion → NO

* Vectors → Component (Break-Karna)
  - addition ✓
  - magnitude ✓

MR* Box → 2-D motion

⟹ 2-D Motion ke question ko Vector ke help se (1-D) & (1-D) me break karo, then separate x-axis & y-axis me solve karo, then again use vector to add them take magnitude of vector

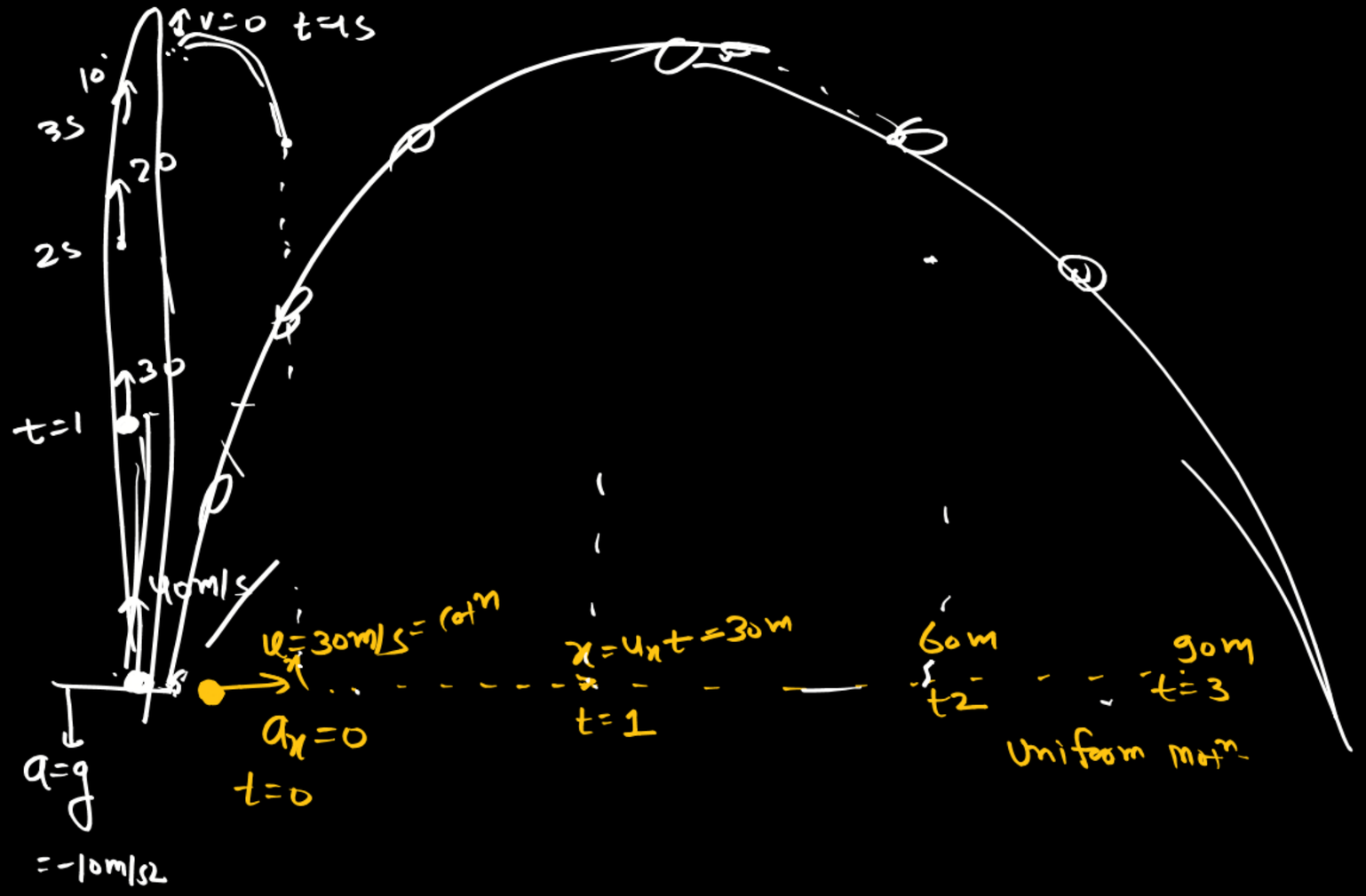
MR* Box

X-axis ka P.Q, x-axis me given P.Q se solve karo. ✓

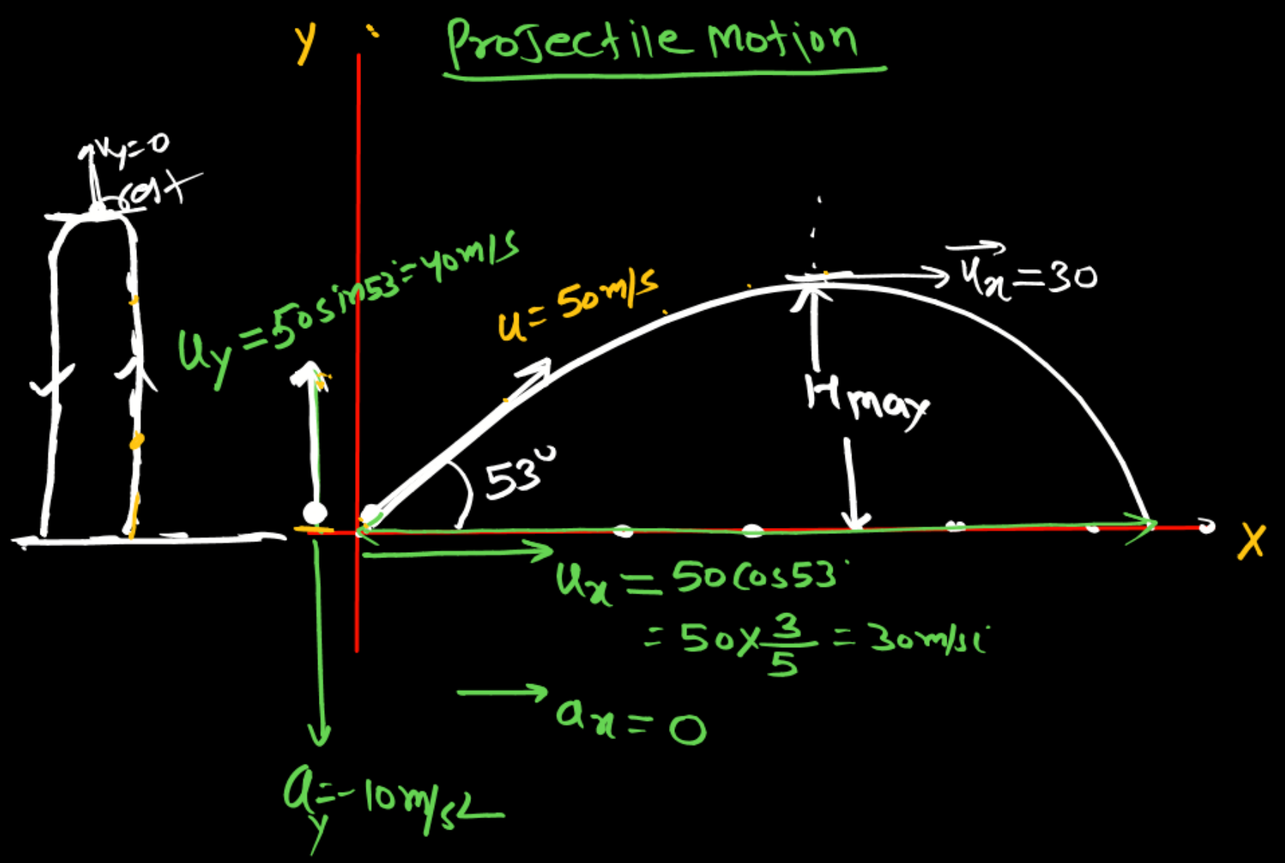
Same in Y

MR* Box

Koin hai jo dono axis ke maje le raha hai → timo


v=0 t=4s
10
35
20
25
30
t=1
40m/s
u=30m/s = const
x=ux t=30m
60m
90m
t=2
t=3
ax=0
t=1
Uniform motn.
a=g
t=0
=-10m/s2

Projectile Motion



X-axis	y-axis	net
$u_x=30$	$u_y=40$	$\vec{u}_{t=0}=30\hat{i}+40\hat{j}$
$a_x=0$	$a_y=-10m/s^2$	$\vec{a}=-g\hat{j}$
$\vec{V}=30m/s$ constant Uniform motion in x	$\vec{v}=u-gt\hat{j}$	$\vec{v}=v_x\hat{i}+v_y\hat{j}$
	$\vec{v}=(40-10t)\hat{j}$	$\vec{r}=x\hat{i}+y\hat{j}$
$x_t=30\times t$	$y=u_yt+\frac{1}{2}a_yt^2$	

① $T_{upward}=\frac{u_y}{g}=\frac{40}{10}=4sec$

② $T_{down}=4sec$

③ $T_{Fly}=8sec.$

④ $H_{max}=\frac{u_y^2}{2g}=\frac{(40)^2}{2\times10}=80m$

⑤ Range $=u_x T_f=30\times8=240m$

⑤ velocity after 1 sec.

$\vec{u}_x=\vec{v}_x=30\hat{i}$ $\quad\vec{v}_y=30\hat{j}$

$\vec{V}=30\hat{i}+30\hat{j}$ $\quad|V|=30\sqrt{2}$

(6) velocity at max^m height.

$\vec{u}_x = 30\ m/s$

(7) disp^m after 1-sec.

$x = u_x t = 30 \times 1 = 30m$

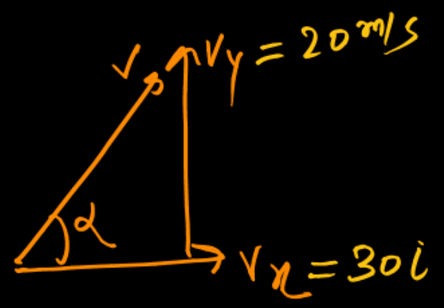
$y = ut + \frac{1}{2}at^2$

$= 40 \times 1 - \frac{1}{2} 10 (1)^2$

$= 40 - 5 = 35m$

$\vec{r} = 30\hat{i} + 35\hat{j}$

(8) find direction of motion at $t = 2sec$.



$\tan\alpha = \frac{20}{30}$

$\tan\alpha = 2/3$

(9) Position at time 't'

$x = 30t$

$y = 40t - \frac{1}{2}10t^2$

$y = 40t - 5t^2$

(10) eq^n of trajectory

$y = 40\left(\frac{x}{30}\right) - 5\left(\frac{x}{30}\right)^2 = \frac{4}{3}x - \frac{5x^2}{900}$

(11) Velocity of collision.

$$\vec{V} = 30\hat{i} - 40\hat{j}$$

(12) Change in velocity in comp^{k} Proton

$$\Delta\vec{V} = \vec{V_f} - \vec{V_i}$$

$$= (30i - 40j) - (30i + 40j)$$

$$= -80j$$

(13) $\vec{u}_{Ay} = \dfrac{\vec{u_i} + \vec{v_f}}{2}$

$= u_n$

(Q) Velocity of Projection $\vec{V} = 40\hat{i} + 30\hat{j}$

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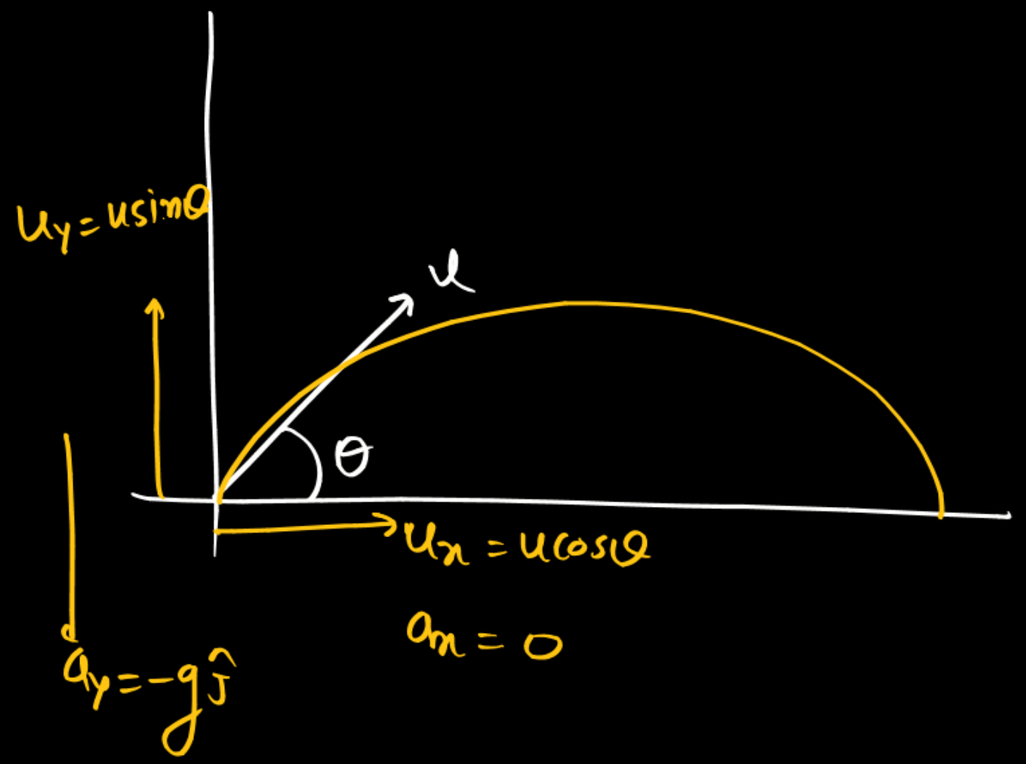
① $T_f = (3+3) = 6\,sec$

② $H_{max} = \frac{30 \times 30}{2 \times 10} = 45m$

③ $Range = u_x T_f = 40 \times 6$
$= 240\,m$

④ $\vec{u}_{t=2sec} = 40\hat{i} + 10\hat{j}$

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$$T_f = \frac{2u\sin\theta}{g}$$

$$H_{max} = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g}$$ ✓

$$R = u_x T_f$$

$$= u\cos\theta\left[\frac{2u\sin\theta}{g}\right]$$ ✓

Question

Velocity of projection $\vec{u} = 40\hat{\imath} + 30\hat{\jmath}$ then find H, T, R and angle of project.

Question

A bullet is fired from a gun at the speed of 280 ms^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is _____.

($g = 9.8\ ms^{-2}$, $\sin 30° = 0.5$) **[NEET-2023]**

1. 80 m
2. 100 m
3. 60 m
4. 40 m

Question

The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is: **[AIPMT Pre. 2012]**

1. $\theta = \tan^{-1}$
2. $\theta = 45°$
3. $\theta = \tan^{-1}\left(\frac{1}{4}\right)$
4. $\theta = \tan^{-1}(4)$

Question

The horizontal range of a projectile is $4\sqrt{3}$ times its maximum height. Its angle of projectile will be:

1. $45°$
2. $60°$
3. $90°$
4. $30°$

Thank

You